# Manish J Bangera

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## Professional Summary

Backend engineer with 4+ years building high-throughput Java services on Kafka, Couchbase, and AWS for a multi-tenant retail platform. Depth in asynchronous execution and retry frameworks, constant-memory streaming over large datasets, and query/index optimization under production load.

## Technical Skills

**Languages:** Java (17/21), Python, SQL
**JVM & Concurrency:** Multithreading, virtual threads, NIO, heap profiling & GC tuning, off-heap/streaming memory patterns
**Distributed Systems:** Apache Kafka (consumer groups, backoff, dead-letter routing), Amazon SQS, Camunda BPMN, idempotency, circuit breakers, distributed transactions
**Data & Storage:** Couchbase (N1QL, covering indexes, MVCC transactions), PostgreSQL, ClickHouse, Qdrant
**Cloud & DevOps:** AWS (S3, Lambda, EventBridge, SQS, EC2, KMS), Kubernetes/EKS, Docker, Jenkins, Gradle
**Testing:** JUnit 5, Mockito, Testcontainers, WireMock

## Experience

**Clover Bay Technologies Pvt Ltd** Bengaluru, India

**Software Development Engineer – 2** Apr 2026 – Present

- **Async Execution Framework** — Published an internal **Spring Kafka library** standardizing consumer config, exponential backoff, and **dead-letter-topic routing**; adopted by 3+ production services, removing per-service retry logic.
- **Product Compatibility Platform** — Designed and shipped a fitment-relationship engine in the PIM, serving compatibility queries across the full catalog.
- **Technical Leadership** — Run system design and code reviews; mentor 2–4 engineers on event-driven patterns.

**Software Development Engineer – 1** Jul 2024 – Mar 2026

- **Constant-Memory Streaming Pipeline** — Rebuilt catalog export on **Apache POI SXSSF/SAX** with paginated **Couchbase** reads, batched **S3** uploads, and async **Kafka** orchestration. Removed a recurring **OOM** failure mode; scaled a 10K-row ceiling to **100K+ rows at flat heap**.
- **Query & Index Optimization** — Cut **P99 latency from 2s+ to sub-second** via **N1QL** execution-plan analysis, covering-index design, predicate rewrites, and **keyset pagination** replacing offset scans.
- **Change-Propagation Connectors** — Built **Shopify** and **MercadoLibre** connectors ingesting webhook events into **Kafka** for near-real-time sync, with ordering and replay guarantees.
- **Stateful Workflow Engine** — Owned the promotions module end to end: domain model, API contracts, and **Spring State Machine** workflows with auditable transitions and **Couchbase transactions** under contention.

**Associate Software Development Engineer** Jul 2022 – Jun 2024

- **Fault-Tolerant Orchestration** — Built a **Camunda**-orchestrated backorder pipeline with dynamic routing, **circuit breakers**, and bounded retries, reaching **99%+ reliability** and removing manual intervention across tenants.
- **Bulk Ingestion** — Designed an event-driven ingestion path on **S3, Lambda, EventBridge, and SQS**, processing 10K-row files in **Kafka** batches with tenant isolation and at-least-once delivery.
- **Testing & Quality** — Introduced **Testcontainers** and **JUnit 5/Mockito** integration testing, raising coverage from 50% to **75%+**.

## Projects

- **Conversational Product Search Agent (POC)** — **LangGraph** agent over a product catalog with **Qdrant** hybrid vector search, **Groq** LLM inference, and multi-turn session state.

## Education

**N.M.A.M Institute of Technology**, B.E. in Computer Science — CGPA: 8.74 / 10 2018 – 2022